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Haowei Lin

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Research Overview

I hope to build next-generation AI with high real-world impact. My current work is centered on **research agents**, where I design self-evolving terminal agents for scientific discovery. Previously I studied **Generative Foundation Models**, where I developed autoregressive and diffusion models across multi-modalities (e.g., image, text, molecules, MDP, 3D).
I am an active contributor to prominent open-source agent projects (e.g., Harbor, Terminal Bench, OpenHands), which have collectively received over 60,000 GitHub stars.

Education

Institude for Artificial Intelligence, Peking University — Ph.D. Student — Sept 2023 – Jul 2028 (expected)

- **Advisors:** Prof. Jianzhu Ma and Prof. Yitao Liang and Prof. Di He

Yuanpei College, Peking University — B.S. in Artificial Intelligence (Honor Track) — Aug 2019 – Jul 2023

- **GPA:** 3.85/4.00 (rank 1st in Major); Top 5% in Yuanpei College (12/305)
- **Bachelor Thesis:** *IneqGPT: A Neural-Symbolic Approach towards Automated Inequality Proving*
- **Research Advisor:** Prof. Bing Liu (Distinguished Professor of UIC)

Selected Publications

(* denotes equal contribution. Refer to Google Scholar for full list.)

Harbor Adapters and Harbor-Index: Infrastructure and a Curated Meta-Dataset for Large-Scale Agentic Evaluation 2026

Lin Shi*, **Haowei Lin***, Zixuan Zhu, Xiaoyue Zhou, Xiang Li, ··· (125 people), Alex Shaw.
Accepted to the Evaluations & Datasets (E&D) Track of **NeurIPS 2026**.

Test-Time Scaling for Scientific Equation Discovery 2026

Haowei Lin*, Hubert Lim*, Xiangyu Wang, Letian Huang, Di He
In The 2026 Conference on Empirical Methods in Natural Language Processing (**EMNLP 2026**).

UniCode: Augmenting Evaluation for Code Reasoning 2026

Xinyue Zheng*, **Haowei Lin***, Shaofei Cai, and Zilong Zheng, Yaodong Yang, Yitao Liang
In The Forty-third International Conference on Machine Learning (**ICML 2026**).

Can Language Models Discover Scaling Laws? 2026

Haowei Lin, Haotian Ye, Wenzheng Feng, Quzhe Huang, Yujun Li, Hubert Lim, Zhengrui Li, Xiangyu Wang, Jianzhu Ma, James Zou, Yitao Liang
In The Fourteenth International Conference on Learning Representations (**ICLR 2026**).

Terminal-Bench: a Benchmark for AI Agents in Terminal Environments 2026

Mike Merrill, Alex Shaw, Nicholas Carlini, Boxuan Li, Harsh Raj, Ivan Bercovich, Lin Shi, Jeong Yeon Shin, Thomas Walshe, E. Kelly Buchanan, Junhong Shen, Guanghao Ye, **Haowei Lin**, ··· (70 people), Ludwig Schmidt
Used by OpenAI, Google DeepMind, Anthropic, Meta, DeepSeek, etc. for LLM evaluation in their model release.
In The Fourteenth International Conference on Learning Representations (**ICLR 2026**).

MCU: An Evaluation Framework for Open-Ended Game Agents 2025

Xinyue Zheng*, **Haowei Lin***, Kaichen He, Zihao Wang, Qiang Fu, Haobo Fu, Zilong Zheng, Yitao Liang.
In The Forty-second International Conference on Machine Learning (**ICML 2025 Spotlight**).

TFG-Flow: Training-free Guidance in Multimodal Generative Flow 2025

Haowei Lin*, Shanda Li*, Haotian Ye, Yiming Yang, Stefano Ermon, Yitao Liang, Jianzhu Ma.
In The Thirteenth International Conference on Learning Representations (**ICLR 2025**).

A Neural Symbolic Model for Space Physics 2025

Jie Ying*, **Haowei Lin***, Chao Yue*, Yajie Chen, Chao Xiao, Quanqi Shi, Yitao Liang, Shing-Tung Yau, Yuan Zhou, Jianzhu Ma.
Nature Machine Intelligence (Cover Paper).

Selecting Large Language Model to Fine-tune via Rectified Scaling Law 2024

Haowei Lin*, Baizhou Huang*, Haotian Ye*, Qinyu Chen, Zihao Wang, Sujian Li, Jianzhu Ma, Xiaojun Wan, James Zou, Yitao Liang.
In The Forty-first International Conference on Machine Learning (**ICML 2024**).

TFG: Unified Training-Free Guidance for Diffusion Models 2024

Haotian Ye*, **Haowei Lin***, Jiaqi Han*, Minkai Xu, Sheng Liu, Yitao Liang, Jianzhu Ma, James Y. Zou, Stefano Ermon.
In Advances in Neural Information Processing Systems 37 (**NeurIPS 2024 Spotlight**).

Class Incremental Learning via Likelihood Ratio-Based Task Prediction 2024

Haowei Lin, Yijia Shao, Weinan Qian, Ningxin Pan, Yiduo Guo, Bing Liu.
In The Twelfth International Conference on Learning Representations (**ICLR 2024**).

Continual Pre-Training of Language Models 2023

Zixuan Ke*, Yijia Shao*, **Haowei Lin***, Tatsuya Konishi, Gyuhak Kim, Bing Liu.
In The Eleventh International Conference on Learning Representations (**ICLR 2023**).

Professional Experience

Tencent Hunyuan, Foundation Model Department — Intern, Agent Team 2026.04 – now

- Lead the end-to-end development of Hy Research Agent, covering evaluation, harness design, training, data scaling, and system iteration.

Harbor x Terminal Bench Team (OSS Project) — Co-lead, Adapters Team 2025.07 – now

- Organize the Adapters Team to adapt 100+ agent benchmarks to Harbor; review pull requests and develop new features to improve agent benchmarking.
- Conduct large-scale benchmarking across 50+ benchmarks and 20+ model-agent combinations; secure funding from model providers and sandbox providers including Anthropic, OpenAI, Google, DeepSeek, MiniMax, Xiaomi, Moonshot, Z.ai, Daytona, Modal, etc.; build experimental infrastructure; and collect over 1M trajectories with more than $50K in funding.

Wizard Intelligence Learning Lab, WizardQuant — Intern, Post-training Team 2025.05 – 2026.04

- Developed novel post-training paradigms for self-evolving agents targeting open scientific problems.
- Designed and implemented a self-evolving agent system from scratch that outperformed GDM's AlphaEvolve in both efficiency and performance for scientific discovery.

DP Technology — Research Intern 2024.12 – 2025.05

- Developed **Uni3DAR**, a unified autoregressive 3D generative model for both understanding and generation.

Helixon — Research Intern 2022.09 – 2023.09

- Worked on multi-conformation protein structure prediction built on AlphaFold.
- Designed peptide-based therapeutics via structure generation and optimization workflows.

Search Group, ByteDance — Summer Intern 2022.06 – 2022.09

- Designed and launched a set of intention analysis models for Douyin Search Engine.

Teaching Assistant 2021 – now

- Spring 2025 & 2024, Introduction to Artificial Intelligence, Peking University.
- Spring 2023, Mathematical Foundation for Artificial Intelligence, Peking University.
- Fall 2022 & 2023 & 2024 & 2025, Natural Language Processing with Deep Learning, Peking University.
- Fall 2022, Introduction to Theoretical Computer Science, Peking University.
- Fall 2021, A Preliminary Seminar on Artificial Intelligence, Peking University.

Conference Reviewer 2022 – now

- ICML: 2022 (**Outstanding Reviewer Award**), 2023, 2024, 2025, 2026 (**Gold Reviewer Award**); NeurIPS: 2022, 2023, 2024, 2025;
  ICLR: 2024, 2025, 2026 (**Top-200 Reviewer Award**); ACL, EMNLP (ARR): 2024, 2025; COLM: 2024, 2025; CVPR: 2024, 2025; AISTATS: 2025.

Awards

- Peking University President Scholarship, 2026
- National Scholarship, 2025
- National Scholarship, 2023
- Outstanding Bachelor Thesis, Peking University

Skills, Interests & Others

Programming Languages: Python, C/C++, LaTeX, Bash, SQL, HTML, CSS, JavaScript

Frameworks & Tools: PyTorch, VeRL, Transformers, Django, Git, Spark, Bootstrap4

Languages: Mandarin (native), English (proficient, TOEFL 105), Hokkien (native), Japanese (JLPT N4-level)

Interests: Singing (*PKU Top-10 singers in 2025*), Guitar, Piano, Painting